

	Covariates	Δelpd	$\text{SE}(\Delta\text{elpd})$	elpd	$\text{SE}(\text{elpd})$	p_{loo}	$\text{se}_{p_{\text{loo}}}$	looic	se_{looic}
IME detect	MLD_vars + time + month	0.00	0.00	-7101.91	49.58	276.47	6.64	14203.83	99.16
	MLD_vars	-227.59	36.22	-7329.51	33.11	270.96	3.14	14659.01	66.21
	time + month	-361.01	39.64	-7462.92	27.69	206.48	2.04	14925.84	55.38
IME strength	MLD_vars + time + month	0.00	0.00	5713.96	79.43	450.04	20.96	-11427.92	158.87
	MLD_vars	-491.53	41.01	5222.43	74.68	486.53	17.23	-10444.86	149.35
	time + month	-793.65	46.33	4920.30	79.15	460.57	20.12	-9840.61	158.31

Table S1 | Model comparison using leave-out-out cross-validation (LOO). Models fitted to probability of IME detection (IME detect; Bernoulli distributed) and IME strength (chl-a enhancement, %; Gamma-distributed), for nested models with smoothers for the mixed layer climatology (monthly mean), anomaly, and long-term trend of time and month (MLD_vars), and temporal smooths (time + month). Top-ranked models have the lowest Bayesian LOO of expected log pointwise predictive density (elpd). Additional estimates provided for effective number of parameters (p_{loo}) and LOO information criterion (looic).